A Mini Project Report

on

"ULTRASONIC RADAR SYSTEM: REAL-TIME OBJECT DETECTION AND DISTANCE MEASUREMENT USING ARDUINO"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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2024-25

DECLARATION

We hereby declare that the work described in this report, entitled **"ULTRASONIC RADAR SYSTEM: REAL-TIME OBJECT DETECTION AND DISTANCE MEASUREMENT USING ARDUINO"** which is being submitted by us in partial fulfillment for the award of the degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women,** affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500085 is the result of original work carried out by us under the guidance of **Dr. Viswanadham Ravuri .**

This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

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Certificate

This is to certify that the mini project report, entitled **"ULTRASONIC RADAR SYSTEM: REAL-TIME OBJECT DETECTION AND DISTANCE MEASUREMENT USING ARDUINO"** is a record of bonafide work carried out by **21WH1A0487 - Y. Sai Ruthvika, 21WH1A04C4 - K. Srinithya, 21WH1A04A0 - B. Leelavanitha, 21WH1A0479 - C. Akhila** in partial fulfillment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women,** affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500085.

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ABSTRACT

The Ultrasonic Radar System is a compact, portable, and cost-effective solution designed for real-time object detection and distance measurement. By integrating an Arduino UNO, ultrasonic sensors, and a servo motor, the system is capable of scanning its surroundings and capturing precise positional data. This data is processed and displayed on a radar-like interface, offering a dynamic and interactive visualization of the detected environment. Its simple yet effective design makes it an ideal choice for various applications.

The primary objective of this project is to develop an affordable system that delivers reliable real-time monitoring capabilities. The Ultrasonic Radar System can detect obstacles, measure distances, and map its surroundings with high accuracy. The radar interface not only enhances user interaction but also provides a clear and intuitive representation of the scanned area. These features allow the system to adapt to diverse scenarios such as surveillance, obstacle detection in autonomous navigation, and security applications.

This approach presents a low-cost, efficient, and adaptable solution with broad usability across multiple fields. Its versatility and scalability make it suitable for both academic and practical purposes, enabling further exploration in the fields of embedded systems, robotics, and automation. With real-time monitoring and precise object detection, the Ultrasonic Radar System provides a strong foundation for addressing modern technological challenges.

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1. INTRODUCTION

Ultrasonic waves, produced by objects through their natural frequency of vibration, interact with the surrounding air and generate sound waves that are often beyond the range of human hearing. Frequencies above 20,000 Hz, known as ultrasonic waves, can be detected using ultrasonic sensors. These sensors incorporate a transducer capable of converting sound energy into electrical energy and vice versa. Ultrasonic sensors are extensively employed in applications such as object position and orientation me11asurement, collision avoidance systems, and surveillance systems.

Ultrasonic technology addresses challenges like linear measurement limitations by enabling non-contact measurements. This capability allows users to measure the distance between objects, their speed, and other parameters with ease. The speed of sound propagation depends on the square root of the ratio between the medium's density and stiffness. Environmental factors, such as temperature, can also influence the speed of sound, further impacting measurements.

The fundamental operation of an ultrasonic sensor involves emitting ultrasonic waves into the air. These waves travel through the medium and reflect back upon encountering an object. By analyzing the properties of the reflected waves, critical information about the object's distance, position, and speed can be obtained. This process is facilitated by a combination of hardware components, such as the ultrasonic sensor, and software tools, including Arduino and processing software.

One of the most common applications of ultrasonic sensors is range finding, often referred to as sonar. Similar to radar systems, sonar involves directing ultrasonic waves in a specific direction. When these waves encounter an object, they reflect back, and the time taken for the waves to return is used to calculate the distance of the object. This principle is utilized by bats in their natural echolocation abilities.

Radar systems, on the other hand, employ radio waves for object detection, enabling the determination of an object's size, height, direction, or speed. Radar systems are designed in various configurations, ranging from compact units for

specific tasks to extensive systems occupying large spaces. Applications of radar extend to air traffic control, long-distance surveillance, missile guidance systems, maritime navigation, and outer space monitoring. Modern radar systems integrate advanced digital signal processing techniques to enhance their functionality and accuracy, supporting critical tasks such as collision avoidance, sea surveillance, and early warning systems.

Both ultrasonic and radar technologies exemplify the power of wave-based detection systems, offering versatile solutions for monitoring, measurement, and safety across a wide range of fields.

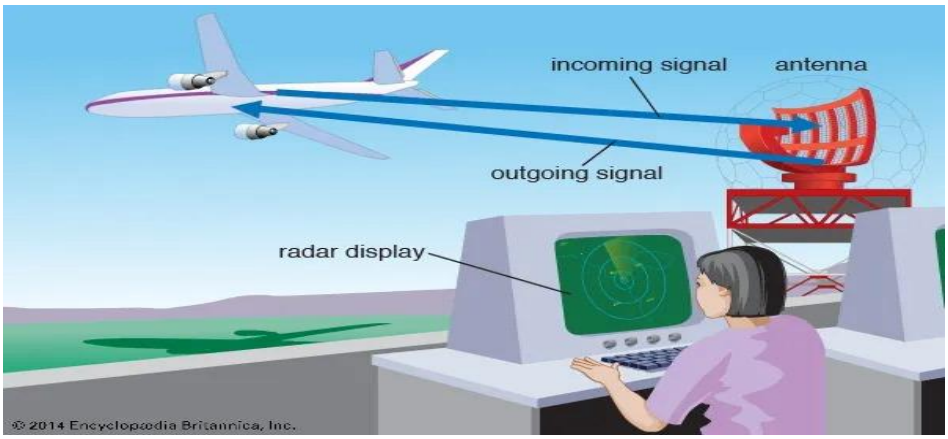

Fig.1: Radar tracking an aircraft using transmitted and reflected signals on a radar screen.

1.1 Objective

The primary objective of the Ultrasonic Radar System project is to design and develop a cost-effective, portable, and reliable solution for real-time object detection and distance measurement. By utilizing an Arduino UNO microcontroller, ultrasonic sensors, and a servo motor, the system aims to achieve precise scanning and visualization of the environment. This objective is driven by the need for affordable and adaptable technologies that can address various challenges in automation, surveillance, and navigation systems. One of the key goals of the project is to provide an interactive radar-like interface that dynamically displays the detected objects and their distances, ensuring an intuitive and user-friendly experience. By making the

technology accessible to professionals, students, and hobbyists, the project opens new opportunities for exploring embedded systems and automation.

Another significant objective is to demonstrate the potential of ultrasonic sensors as a reliable and low-cost alternative to expensive detection systems like LIDAR. The project explores how ultrasonic sensors, combined with servo motors, can enhance efficiency and range through angular scanning, showcasing their practical applications in real-world scenarios. Additionally, the project highlights the versatility of the Ultrasonic Radar System across multiple domains, contributing to advancements in embedded machine learning, robotics, and automation. By focusing on real-time monitoring, precision, and adaptability, the system addresses the growing demand for intelligent, low-cost solutions in modern automation and environmental sensing technologies.

1.2. Scope of the Project

The Ultrasonic Radar System offers a versatile and cost-effective solution for real-time object detection and distance measurement. The project has broad applicability and potential for further development, with the following scopes:

1. Real-Time Object Detection and Distance Measurement

The system provides highly accurate real-time object detection and distance measurement, making it an essential tool in various applications such as:

- Autonomous Vehicles: Enhances the safety and efficiency of self-driving cars by detecting obstacles in real-time.
- Industrial Automation: Improves operational accuracy in automated production lines by detecting objects and ensuring seamless processes.

2. Surveillance and Security

The radar system can be strategically deployed in security-sensitive areas to:

- Detect and track intrusions.
- Monitor unauthorized access, providing real-time alerts to enhance safety and security. Its reliable performance ensures a robust surveillance framework for homes, offices, and industrial premises.

3.Obstacle Avoidance for Autonomous Robots

By integrating the system into autonomous robots, it enables:

- Effective obstacle detection and avoidance in real-time.
- Smooth and safe navigation in dynamic and unpredictable environments such as warehouses, hospitals, and disaster zones.

4.Customizable and Scalable Design

The system's modular design supports:

- Easy customization to meet specific project requirements.
- Scalability for integration with advanced technologies like:
    - Machine learning for predictive analytics.
    - Wireless communication for remote monitoring and control.
    - IoT for enhanced connectivity and data exchange across devices.

5.Low-CostImplementation

Using affordable components such as Arduino UNO, ultrasonic sensors, and servo motors, the system:

- Ensures accessibility for a wide range of users, including students, researchers, and developers.
- Can be easily replicated in resource-constrained environments without compromising performance.

2. LITERATURE SURVEY

Ultrasonic radar systems, especially when paired with microcontrollers like Arduino, have become increasingly popular due to their affordability, simplicity, and versatility. These systems work by emitting ultrasonic waves that reflect off objects and return to the sensor, enabling real-time object detection and distance measurement. While research on these systems began in the early 2000s, their widespread use in applications like surveillance, security, and autonomous navigation grew significantly after 2010. This surge in adoption is largely due to the availability of low-cost microcontrollers and sensors, making them accessible for both academic and practical purposes.

In the realm of surveillance and security, ultrasonic radar systems have been studied and deployed extensively since the mid-2010s. Researchers focused on integrating these systems with motion detection technologies, improving tracking and environmental awareness. Notable studies in 2016 and 2017 highlighted the effectiveness of ultrasonic systems in perimeter monitoring, particularly for cost-effective and easy-to-install solutions suitable for various applications. By combining ultrasonic sensors with servo motors, these systems gained broader scanning capabilities, allowing for dynamic environmental coverage and enhanced real-time object tracking. This development made them a popular choice for enhancing safety in security-sensitive areas.

Ultrasonic radar systems have also been critical in autonomous navigation, particularly for robots and drones. Since the early 2010s, these systems have been used for obstacle detection and pathfinding, offering a cost-effective alternative to expensive LiDAR systems. Between 2015 and 2020, research demonstrated that ultrasonic sensors could perform mapping and obstacle detection tasks effectively. However, challenges such as sensor limitations, environmental noise, and slow scanning speeds have prompted further research. From 2018 onward, improvements in sensor hardware and data processing algorithms have addressed some of these issues, boosting the range, accuracy, and overall performance of ultrasonic systems.

Recent studies, particularly from 2021 and 2022, have explored ways to enhance the reliability and accuracy of ultrasonic radar systems by integrating them with other sensors, such as infrared or PIR sensors. These combinations have proven useful in improving detection accuracy, reducing false positives, and enhancing performance in complex environments. Despite these advancements, challenges like reflective surfaces and ambient noise continue to affect the systems, leading to ongoing research into better signal processing techniques to overcome these obstacles.

The scalability and simplicity of ultrasonic radar systems have led to their adoption across diverse fields. Research has increasingly focused on integrating IoT technologies for remote monitoring and control, broadening their potential applications to areas like disaster management and agriculture. Additionally, advancements in data visualization, such as real-time object classification using machine learning, have made these systems more effective for smart surveillance and intelligent robotics. With continuous improvements in sensor technology, data algorithms, and the integration of wireless communication and AI, ultrasonic radar systems hold great promise for future applications in both research and industry.

3. SYSTEM DESIGN AND METHODOLOGY

The design and architecture of an ultrasonic radar system using Arduino involve several key components working together to detect and visualize objects in real-time. The core of the system includes an Arduino UNO microcontroller, which controls the operation of ultrasonic sensors, typically mounted on a servo motor for scanning the environment. The ultrasonic sensors emit sound waves and measure the time it takes for them to bounce back after hitting an object, allowing the system to calculate distances. The servo motor rotates the sensors to cover a wide area, enabling dynamic scanning. The data collected is processed by the Arduino and displayed on a user interface, often a radar-like visualization, providing a clear view of object positions and distances. The system is powered by a battery or an external power supply, and it can be customized with additional sensors or communication modules for enhanced functionality in specific applications such as security, robotics, or autonomous navigation.

3.1 Hardware Equipment

The conceptual design focuses on making the game accessible by incorporating auditory feedback, enhancing the gameplay for visually impaired users. The core functionality of the game includes an electronic board, dice with speech output, and an interface that delivers real-time updates.

3.1.1 Arduino UNO

Arduino Uno is powered by the ATmega328P microcontroller, which has 32 KB of flash memory, 2 KB of SRAM, and 1 KB of EEPROM. Acts as the central processing unit for data acquisition and control.The Arduino Uno serves as the core controller for an ultrasonic radar system. It processes data received from the ultrasonic sensor, which measures the distance to obstacles by emitting and receiving ultrasonic waves. The Uno’s digital I/O pins facilitate communication with the sensor and actuators, while its analog inputs handle additional sensor data if needed. The Arduino reads the time-of-flight data from the ultrasonic sensor, calculates the distance to an object, and updates the radar’s visual interface.

Fig.2: Arduino UNO

3.1.2 USB Cable

The USB cable provides power to the Arduino Uno and facilitates communication between the Arduino and a computer for programming and data exchange. The USB cable serves a dual purpose in the ultrasonic radar system: powering the Arduino Uno and allowing the user to upload the radar program to the microcontroller. During development, it enables real-time data monitoring and debugging by maintaining a continuous connection between the Arduino and the computer. Its role in programming and powering the system makes it indispensable for initializing and testing the radar functionality.

Fig.3: USB Cable for Arduino Uno

3.1.3 Ultrasonic Sensor (e.g., HC-SR04)

Measures the distance and position of objects by emitting ultrasonic waves and receiving the reflected waves. The ultrasonic sensor acts as the primary distance-measuring component in the radar system. It emits ultrasonic pulses and measures the time taken for the reflected echoes to return, which allows the system to calculate the distance to nearby objects. When integrated with a servo motor, the sensor can rotate, enabling it to scan across a wide field of view. The sensor's data is then processed by the Arduino, which plots the detected objects on a visual interface, simulating a radar display. Its reliable operation in various environmental conditions and non-contact sensing capability make it an essential component for creating an effective ultrasonic radar system.

Fig.4: Ultrasonic Sensor

The working principle of ultrasonic sensor is as follows:

High level signal is sent for 10us using Trigger. The module sends eight 40 KHz signals automatically, and then detects whether pulse is received or not. If the signal is received, then it is through high level. The time of high duration is the time gap between sending and receiving the signal.

Distance= (Time x Speed of Sound in Air (340 m/s))/2

3.1.4 Servo Motor

Servo motors provide precise control over angular rotation, typically between 0° and 180°, making them ideal for applications requiring accurate positioning. The servo motor is responsible for enabling the ultrasonic sensor to rotate and scan its surroundings, allowing the radar system to cover a wide field of view. Controlled by the Arduino, the servo motor adjusts the angle of the ultrasonic sensor step by step, ensuring systematic and precise scanning. The motor's angular position directly determines the direction in which the sensor measures distances, which is crucial for plotting an accurate radar map. Its smooth operation and positional accuracy make it a vital component for simulating real-time radar functionality.

Fig.5: Servo Motor

3.1.5 Breadboard

A breadboard is a reusable platform for assembling and testing electronic circuits without soldering, making it an essential tool for prototyping and experimentation. The breadboard acts as the central platform for connecting and organizing the components of the ultrasonic radar system, such as the Arduino, ultrasonic sensor, and servo motor. The breadboard's modularity allows quick adjustments to the circuit, enabling seamless integration and troubleshooting of the radar system. Its flexibility and reusability make it an invaluable component in building and refining the system.

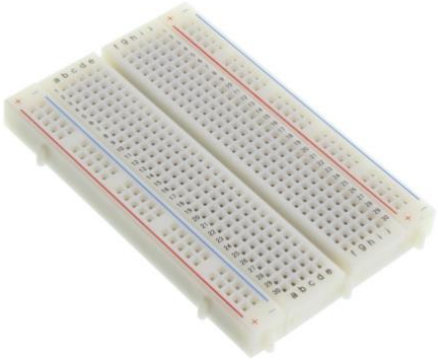

Fig.6: Breadboard

3.1.6 Jumper Wires

Jumper wires are essential for connecting the various components of the ultrasonic radar system, including the Arduino, ultrasonic sensor, and servomotor motor.

Fig.7: Jumper Wires

3.2 Software Equipment

3.2.1 Arduino IDE

The Arduino IDE provides a user-friendly interface to write, compile, and upload code to the Arduino Uno for controlling the ultrasonic radar system. It includes built-in libraries, such as Servo.h for controlling the servo motor, and allows importing third-party libraries for enhanced functionalities, streamlining development for the radar system.

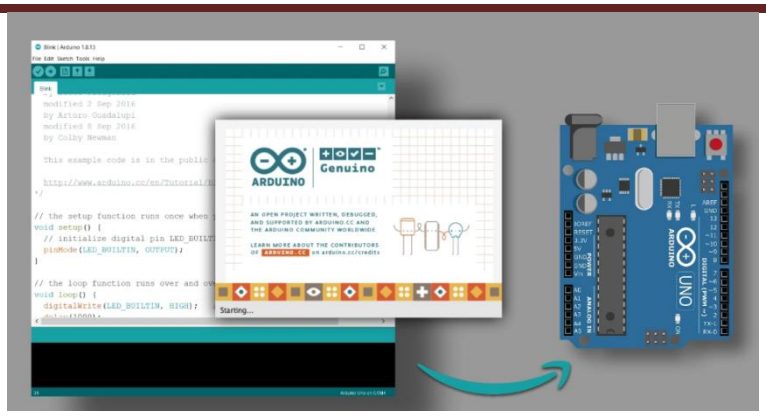

Fig.8: Arduino IDE Software

3.2.2 Processing IDE

Processing IDE is a flexible software sketchbook and language used for visual arts and design, often applied to create visualizations and interactive applications. Visualizing the data from the Arduino as a real-time radar interface. Receiving data from the Arduino via serial communication and translating it into a user-friendly graphical interface.

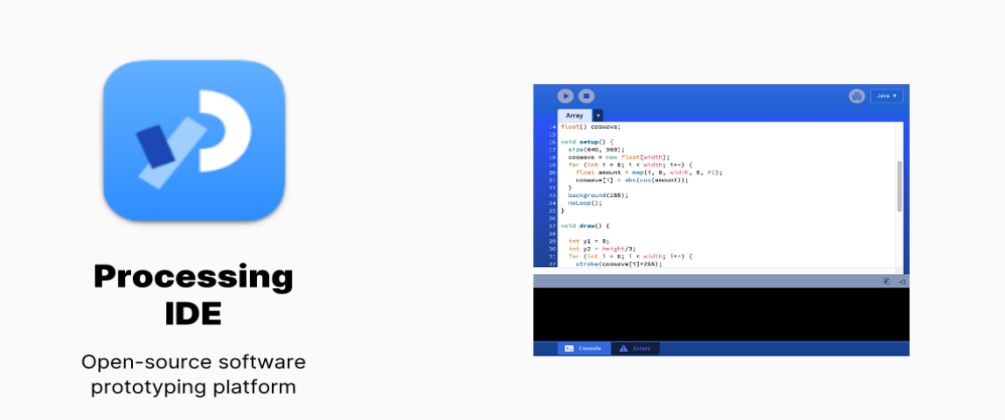

Fig 9 : Processing IDE Software

Specifications

- Free to download and open source
- Interactive programs with 2D, 3D or PDF output
- OpenGL integration for accelerated 2D and 3D
- For GNU/Linux, Mac OS X, and Windows
- Over 100 libraries extend the core software

3.2.3 Jupyter Notebook

Jupyter Notebook is an open-source interactive computing environment that allows developers to create and share documents containing live code, equations, visualizations, and explanatory text. In this project, Jupyter Notebook is utilized to process and visualize data received from the Arduino. Reads real-time data (distance and angle) from the Arduino via serial communication. Creates a radar-like plot to represent the detected objects dynamically.

Fig 10 : Jupyter Python Notebook

3.3 Methodology

In order to testify the working of this system, after its designing, construction and programming we placed few objects in front of the ultrasonic sensor. As the motor started to rotate, our monitor started to display the output through processing IDE. Hence, when the sensor crossed over the object it showed a red segment with the distance and angle where the object is placed.

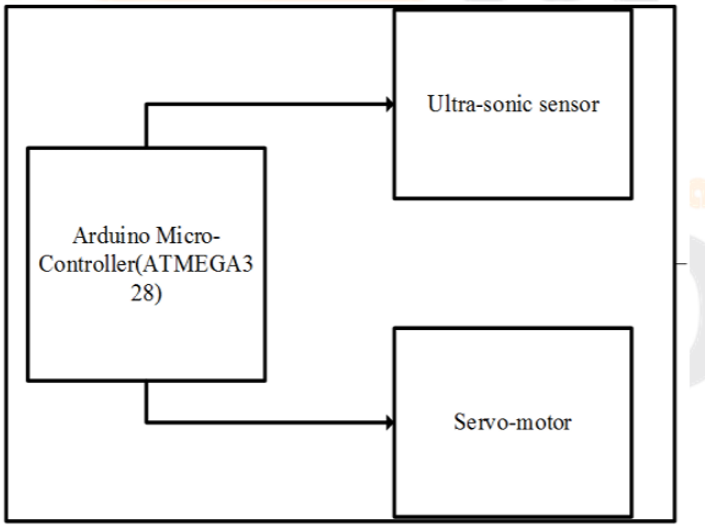

Fig11: Hardware System Design of Radar System

The above figure represents a brief overview of this radar system. Here, as it is shown the controller we are using is Arduino, with the input Ultrasonic sensor and the output is the servo motor which rotates 180 degrees. The microcontroller controls all the operations of this system, from rotation of the motors to the object detection of the ultrasonic and representation of the result on the screen.

Here, it can be seen how the work flow in this radar system. The sensor is going to sense the obstacle and determine the angle of incident and its distance from the radar. The servo motor is constantly rotating to and fro, hence making the sensor move. The data obtained is encoded and fed to the processing IDE which represents it on the screen. The results are displayed further in this paper. All these operation are done by Arduino microcontroller from the rotation of the servo, data collection from the sensor, feeding the data to encoder to transferring it to the display.

3.3.1 Working

The basic objective of our design is to ascertain the distance position of the object set at some distance from the sensor. Ultrasonic sensor sends the ultrasonic wave in various ways by rotating with help of servo motors. This wave goes in air and gets reflected back subsequent to striking some object. This wave is again detected by the sensor and its qualities is analyzed and output is shown in screen indicating parameters, for example, distance and position of object. Arduino IDE is utilized to compose code

and transfer coding in Arduino and causes us to detect position or angle of servo motor and it is communicated through the serial port alongside the covered distance of the nearest object in its way. Output of all of this working is shown in the software called processing, it will display the input/output and the range of the object. Implementations of the sensors are done in such a way that ultra-sonic sensor is attached on top of the servo motor because it have to detect the object and its distance. Arduino (micro-controller) will control the ultra-sonic sensor and servo motor and also powered will be given to both of them through micro-controller.

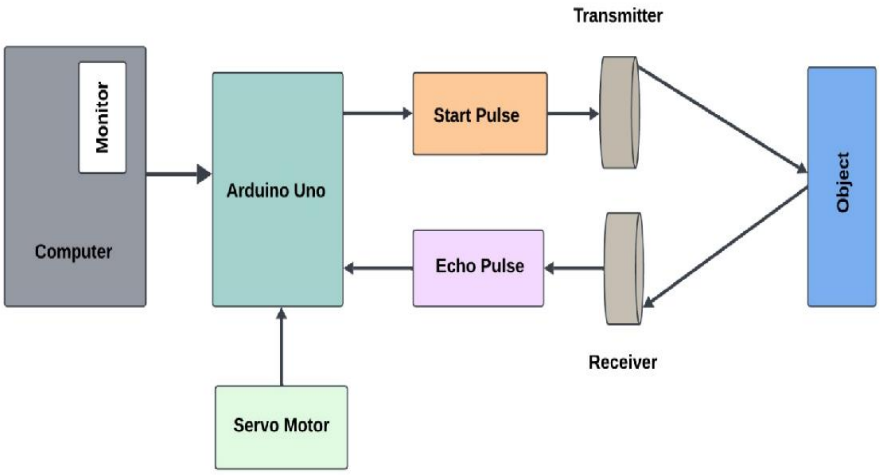

Fig.12 : Ultrasonic Radar System Block Diagram

4.IMPLEMENTATION

To implement the Ultrasonic Radar System, begin by connecting an ultrasonic sensor (e.g., HC-SR04) to an Arduino UNO to measure distances and detect objects. Mount the sensor on a servo motor to enable rotational scanning, covering a predefined angular range. Use a simple serial communication protocol to send the processed data to a computer or display module. Power the system using a stable 5V supply and ensure proper wiring for reliability. Finally, test and calibrate the system to optimize performance for specific applications like surveillance, navigation, or object detection.

4.1 Hardware Integration

Hardware integration for the Ultrasonic Radar System involves connecting key components such as the Arduino UNO, ultrasonic sensor, servo motor, and power supply. Start by wiring the ultrasonic sensor (e.g., HC-SR04) to the Arduino's digital input and output pins for triggering and receiving echo signals. Mount the sensor onto a servo motor, which will allow rotational movement for scanning the environment. Connect the servo motor's signal pin to one of the Arduino's PWM-capable pins to control its rotation. Ensure that all components are powered correctly using either a USB connection or an external 5V power source, taking care to prevent overloading the Arduino's onboard regulator. Use a breadboard for prototyping and testing the connections, ensuring all wires are securely connected to avoid interruptions during operation.

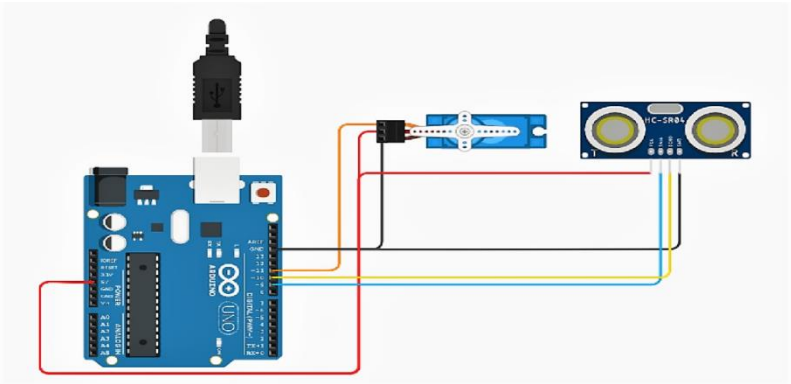

Fig.13: Hardware System Design

To integrate the system's output, connect the Arduino to a display module or a computer via USB. The Arduino can send data to a PC where a graphical user interface (GUI) visualizes the radar output. The GUI can be created using tools like Processing or Python, allowing a dynamic radar-like visualization of the scanned objects. Additionally, ensure the mechanical setup is stable, with the ultrasonic sensor and servo motor mounted on a sturdy base for smooth operation. Finally, test the integrated system by rotating the servo motor and capturing distance data, fine-tuning the connections and configurations to ensure accurate and reliable performance.

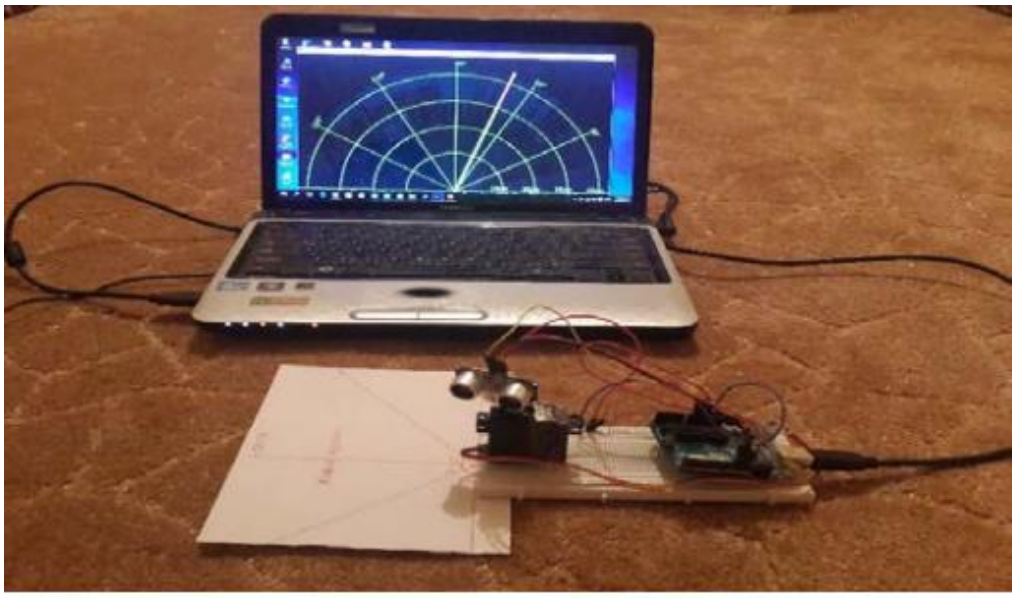

Fig 14 : Hardware Model of the project

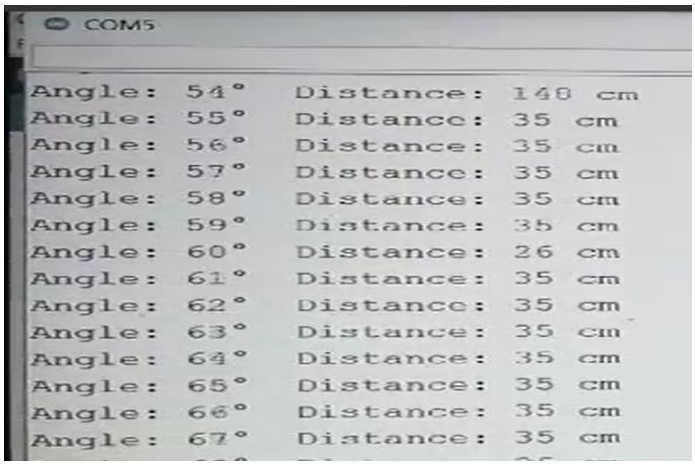

Fig.15 : Serial Monitor Display in Arduino IDE

4.2 Software Development

The software development for the Ultrasonic Radar System begins with programming the Arduino UNO to control the ultrasonic sensor and servo motor. Using the Arduino IDE, write a sketch that sends trigger signals to the ultrasonic sensor and measures the time it takes for the echo to return, calculating the distance to objects based on the time-of-flight formula. Simultaneously, program the servo motor to rotate incrementally across the scanning range (e.g., 0° to 180°) using PWM signals. At each step, capture the distance data from the ultrasonic sensor and the corresponding servo angle, storing this information in variables. Include calibration routines to ensure accurate measurements, considering factors such as sensor delays and environmental conditions.

For visualization, the Arduino should send the distance and angle data to a connected computer or display module via serial communication. The GUI should read the serial data, process it, and plot the detected objects on a radar-like interface, simulating the scanning process in real-time. Add features such as real-time updates, object detection markers, and a dynamic range display to enhance usability. Additionally, ensure the software includes error handling for sensor anomalies and serial communication failures. Test the system thoroughly by scanning various environments, refining the code for smoother performance and better adaptability across different applications.

5. RESULTS

The Ultrasonic Radar System successfully detects objects and visualizes their positions and distances in real-time. The key observations and outcomes from the system are as follows:

5.1 Radar Interface Visualization

The radar-like interface dynamically plots detected objects based on their distance and angle. The visualization provides a clear representation of the surrounding environment, with detected objects highlighted in the radar sweep.

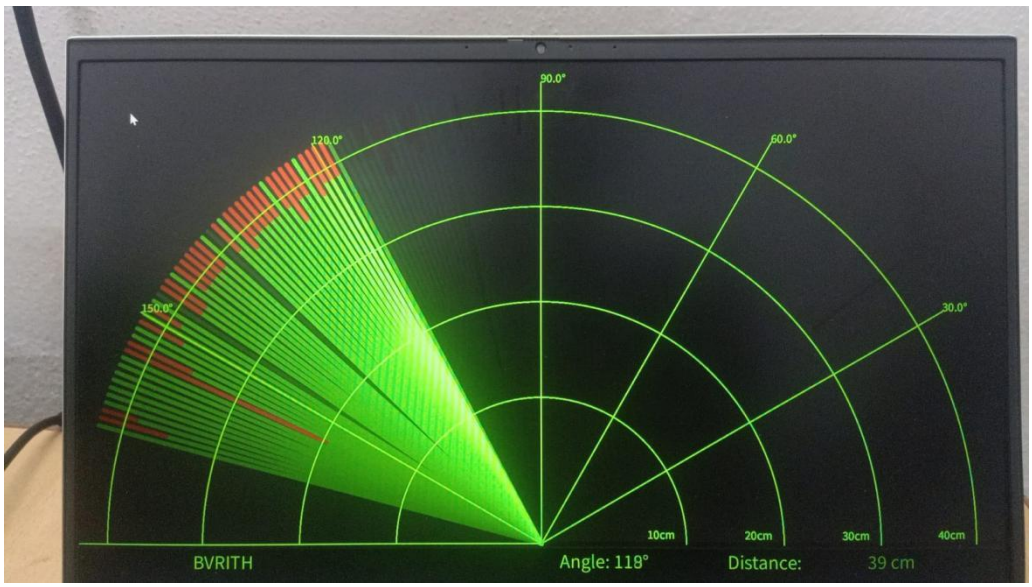

Fig.16: Radar System Visualization

5.2 Distance and Angle Measurement

Using a machine learning model (k-Nearest Neighbors) in Python, the system successfully classifies detected objects based on their distance values. For example, objects such as "Door" and "Table" were identified, demonstrating the system's ability to label objects intelligently.

jupyter Untitled4 Last Checkpoint: 12/23/2024 (autosaved)

In [43]:
```
#Main code for distance and angle measure
import serial
import time

# Initialize Serial Communication
arduino = serial.Serial(port='COM6', baudrate=9600, timeout=1)  # Replace 'COM3' with your Arduino port

def read_data():
    while True:
        if arduino.in_waiting > 0:
            data = arduino.readline().decode('utf-8').strip()
            print(data)  # Print data received from Arduino

# Call the function
read_data()
```

```
Angle: 15°  Distance: 9 cm
Angle: 16°  Distance: 8 cm
Angle: 17°  Distance: 26 cm
Angle: 18°  Distance: 8 cm
Angle: 19°  Distance: 9 cm
Angle: 20°  Distance: 9 cm
Angle: 21°  Distance: 9 cm
Angle: 22°  Distance: 9 cm
Angle: 23°  Distance: 9 cm
Angle: 24°  Distance: 8 cm
Angle: 25°  Distance: 8 cm
Angle: 26°  Distance: 9 cm
Angle: 27°  Distance: 8 cm
Angle: 28°  Distance: 9 cm
```

Fig.17: Distance and angle measurements

5.3 Real-Time Object Detection Results

The real-time object detection results are displayed through a Processing interface, showing the detected object's position and distance. The system continuously updates the data as the ultrasonic sensor scans, providing dynamic feedback on object locations. This allows for efficient monitoring and interpretation of the surrounding environment.

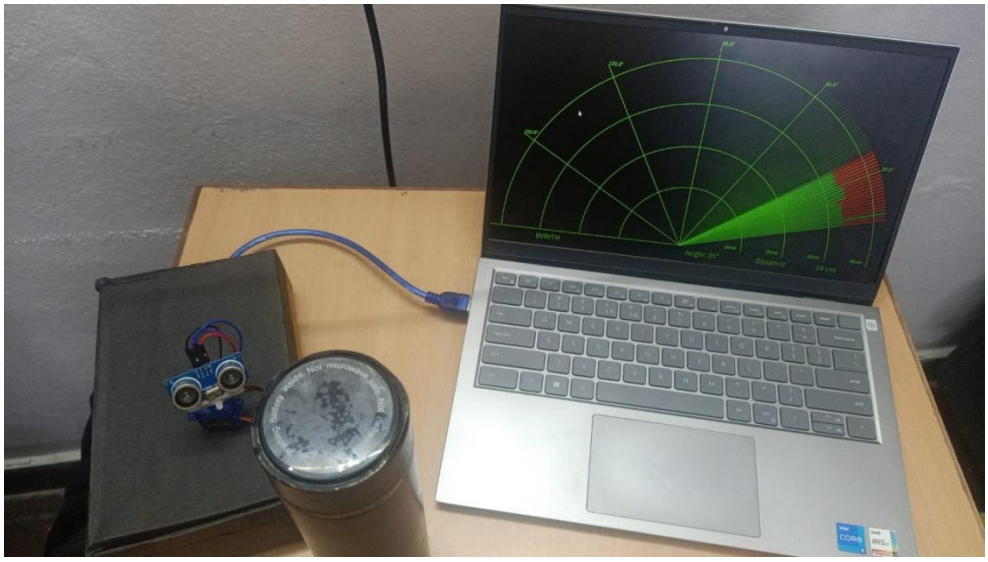

Fig.18: Radar with object

```
In [45]: #Labeling the object
from sklearn.neighbors import KNeighborsClassifier
import numpy as np

# Training data: [distance (cm)] and labels
X_train = [[10], [30], [50], [70], [120], [150], [40], [90], [200]]
y_train = ['Wall', 'Person', 'Door', 'Speed Breaker', 'Vehicle', 'Animal',
           'Table', 'Trash', 'Tree']

# Train KNN model
knn = KNeighborsClassifier(n_neighbors=1)  # Using 1-nearest neighbor for simplicity
knn.fit(X_train, y_train)

# Classify new data
def classify_object(distance):
    label = knn.predict([[distance]])
    return label[0]

# Example usage
distance = 45  # Simulated or real distance
print(f"Object Detected: {classify_object(distance)}")

Object Detected: Door
```

Fig.19: Object Classification

6. CONCLUSION AND FUTURE SCOPE

The ultrasonic radar system has proven to be a reliable, efficient, and cost-effective solution for real-time object detection and classification. By integrating Arduino-based hardware with Python-based software, it offers a blend of simplicity and functionality. The radar interface enhances user interaction and visualization, while the machine learning classification system provides additional insights into the detected environment. These features make it ideal for applications such as surveillance, autonomous navigation, and security systems.

Future scope includes enhancing object classification by incorporating more advanced machine learning models, like neural networks, to improve accuracy and handle a wider range of objects. Another focus is increasing the range and resolution of the ultrasonic sensor for better accuracy in larger areas. Integration with IoT technologies is also planned to enable remote monitoring and control, making the system more versatile and accessible.

The system can also adapt to different environmental conditions by adding sensors such as temperature and humidity. Scalability and mobility will be explored by expanding the system for mobile platforms like robots or drones, enabling autonomous navigation and exploration. Implementing 3D mapping with multi-axis rotation of the sensor could also improve applications in robotics and advanced surveillance systems.

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8. Appendices

Appendix A : Arduino Code:

```
#include <Servo.h>
#define trigPin 8
#define echoPin 9
long duration;
int distance;
Servo myservo;
// Function to calculate distance using the ultrasonic sensor
int calculateDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2; // Convert time to distance in cm
  return distance;
}
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  myservo.attach(11); // Attach the servo to pin 11
```

```
  Serial.begin(9600); // Begin serial communication
}
void loop() {
  int i;
  // Sweep the servo from 15° to 165°
  for (i = 15; i <= 165; i++) {
    myservo.write(i);  // Move the servo to the current angle
    delay(15);         // Wait for the servo to reach its position
    calculateDistance();
    // Print angle and distance with labels
    Serial.print("Angle: ");
    Serial.print(i);
    Serial.print("°  Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
  }
  // Sweep the servo back from 165° to 15°
  for (i = 165; i >= 15; i--) {
    myservo.write(i);  // Move the servo to the current angle
    delay(15);         // Wait for the servo to reach its position
    calculateDistance();
    // Print angle and distance with labels
    Serial.print("Angle: ");
    Serial.print(i);
    Serial.print("°  Distance: ");
    Serial.print(distance);
```

```
    Serial.println(" cm");

  }
}
```

Appendix B : Processing IDE Code

```
import processing.serial.*; // Library for serial communication
import java.awt.event.KeyEvent; // Library for handling key events
import java.io.IOException;

Serial myPort; // Serial communication object

// Variables
String angle = "";
String distance = "";
String data = "";
String noObject;
float pixsDistance;
int iAngle, iDistance;
int index1 = 0;
int index2 = 0;

void setup() {
  //size(1200, 700); // Screen size (Adjust as needed)
  fullScreen();
  smooth();
  myPort = new Serial(this, "COM6", 9600); // Initialize serial communication
  myPort.bufferUntil('.'); // Read data until '.' character
}

void draw() {
  // Background fade effect
  noStroke();
  fill(0, 4);
  rect(0, 0, width, height - height * 0.065);

  fill(98, 245, 31); // Green color
  drawRadar(); // Draw radar grid
  drawLine(); // Draw radar sweeping line
  drawObject(); // Plot detected object
  drawText(); // Display text information
}
```

```
// Handles serial input
void serialEvent(Serial myPort) {
  data = myPort.readStringUntil('.'); // Read data up to '.'
  data = data.substring(0, data.length() - 1);

  index1 = data.indexOf(","); // Find index of ','
  angle = data.substring(0, index1); // Extract angle
  distance = data.substring(index1 + 1); // Extract distance

  iAngle = int(angle); // Convert angle to integer
  iDistance = int(distance); // Convert distance to integer
}

// Draw the radar grid
void drawRadar() {
  pushMatrix();
  translate(width / 2, height - height * 0.074); // Set origin
  noFill();
  strokeWeight(2);
  stroke(98, 245, 31);

  // Draw concentric arcs
  arc(0, 0, (width - width * 0.0625), (width - width * 0.0625), PI, TWO_PI);
  arc(0, 0, (width - width * 0.27), (width - width * 0.27), PI, TWO_PI);
  arc(0, 0, (width - width * 0.479), (width - width * 0.479), PI, TWO_PI);
  arc(0, 0, (width - width * 0.687), (width - width * 0.687), PI, TWO_PI);

  // Draw angle lines
  for (int angle = 0; angle <= 150; angle += 30) {
    line(0, 0, (-width / 2) * cos(radians(angle)), (-width / 2) * sin(radians(angle)));
  }
  popMatrix();
}

// Draw the detected object
void drawObject() {
  pushMatrix();
  translate(width / 2, height - height * 0.074); // Set origin
  strokeWeight(9);
  stroke(255, 10, 10); // Red color
  pixsDistance = iDistance * ((height - height * 0.1666) * 0.025);
```

```
  if (iDistance < 40) { // Plot object if in range
    line(pixsDistance * cos(radians(iAngle)), -pixsDistance * sin(radians(iAngle)),
         (width - width * 0.505) * cos(radians(iAngle)),
         -(width - width * 0.505) * sin(radians(iAngle)));
  }
  popMatrix();
}

// Draw the radar's sweeping line
void drawLine() {
  pushMatrix();
  strokeWeight(9);
  stroke(30, 250, 60); // Green color
  translate(width / 2, height - height * 0.074); // Set origin
  line(0, 0, (height - height * 0.12) * cos(radians(iAngle)),
       -(height - height * 0.12) * sin(radians(iAngle)));
  popMatrix();
}

// Display text and labels
void drawText() {
  pushMatrix();
  if (iDistance > 40) {
    noObject = "Out of Range";
  } else {
    noObject = "In Range";
  }

  fill(0, 0, 0);
  noStroke();
  rect(0, height - height * 0.0648, width, height);
  fill(98, 245, 31);
  textSize(25);

  // Distance markers
  text("10cm", width - width * 0.3854, height - height * 0.0833);
  text("20cm", width - width * 0.281, height - height * 0.0833);
  text("30cm", width - width * 0.177, height - height * 0.0833);
  text("40cm", width - width * 0.0729, height - height * 0.0833);

  // Information display
```

```
  textSize(40);
  text("BVRITH", width - width * 0.875, height - height * 0.0277);
  text("Angle: " + iAngle + "°", width - width * 0.48, height - height * 0.0277);
  text("Distance: ", width - width * 0.30, height - height * 0.0277);
  if (iDistance < 40) {
    text(iDistance + " cm", width - width * 0.15  , height - height * 0.0277);
  }

  // Angle labels
  textSize(25);
  float[] angles = {30, 60, 90, 120, 150};
  for (float angle : angles) {
    float x = width / 2 * cos(radians(angle));
    float y = -width / 2 * sin(radians(angle));
    text(angle + "°", width / 2 + x, height - height * 0.074 + y);
  }
  popMatrix();
}
```

Appendix C : Python Code

```
#Main code for distance and angle measure

import serial

import time

# Initialize Serial Communication

arduino = serial.Serial(port='COM6', baudrate=9600, timeout=1)  # Replace 'COM3'
with your Arduino port

def read_data():

    while True:

        if arduino.in_waiting > 0:

            data = arduino.readline().decode('utf-8').strip()

            print(data)  # Print data received from Arduino

# Call the function

read_data()

#Labeling the object
```

```
from sklearn.neighbors import KNeighborsClassifier

import numpy as np

# Training data: [distance (cm)] and labels

X_train = [[10], [30], [50], [70], [120], [150], [40], [90], [200]]

y_train = ['Wall', 'Person', 'Door', 'Speed Breaker', 'Vehicle', 'Animal',

           'Table', 'Trash', 'Tree']

# Train KNN model

knn = KNeighborsClassifier(n_neighbors=1)  # Using 1-nearest neighbor for
simplicity

knn.fit(X_train, y_train)

# Classify new data

def classify_object(distance):

    label = knn.predict([[distance]])

    return label[0]

# Example usage

distance = 45  # Simulated or real distance

print(f"Object Detected: {classify_object(distance)}")

#labeling object2

from sklearn.neighbors import KNeighborsClassifier

import numpy as np

# Training data: [distance (cm)] and labels

X_train = [[15], [50], [120]]  # Example distances

y_train = ['Wall', 'Person', 'Open Space']

# Train KNN model

knn = KNeighborsClassifier(n_neighbors=1)

knn.fit(X_train, y_train)
```

```
# Classify new data
def classify_object(distance):
    label = knn.predict([[distance]])
    return label[0]
# Example usage
distance = 45  # Simulated or real distance
print(f"Object Detected: {classify_object(distance)}")
```